

Model Evasion, Limits, and Recommendations

Systems Can Be Fooled

**LIMITS
MATTER**

Today we use evasion thinking to make final project recommendations more realistic and limitations more honest.

Due May 26, 2026

Plain-language definition

Evasion means avoiding the rule, signal, or detector

A system can miss what it was supposed to catch.

- Attackers adapt to controls
- Normal behavior can look suspicious
- Suspicious behavior can look normal

Two useful terms

False positives and false negatives

False positive

Something normal gets treated as risky.

False negative

Something risky gets treated as normal.

Tradeoff

Changing a rule can reduce one error while increasing another.

Thresholds

Rules create boundaries, and boundaries create tradeoffs

- Thirty failed logins might trigger review
- Five MFA denials might trigger review
- High severity might trigger escalation
- Every threshold can miss context

Threat model

A threat model asks what can go wrong

1

What needs protection?

2

Who or what could cause harm?

3

What weakness could be exploited?

4

Who would be affected?

5

What control could reduce the risk?

Coding path

For coding projects: your score is not a verdict

- A score can rank systems for review
- A score cannot prove compromise by itself
- A chart can show a pattern
- A chart cannot explain every cause

Essay path

For MLA essays: a strong claim still needs limits

- Do not turn one case into every case
- Do not treat one source as total proof
- Admit what your evidence cannot show
- Use limits to make the recommendation realistic

Recommendations

A realistic recommendation has a specific action

Avoid: improve security.

- Name the action
- Name the evidence
- Name the benefit
- Name the privacy or ethics tradeoff

Limitation paragraph

A limitation tells readers what not to assume

- What your project can show
- What your project cannot prove
- Why that limit matters
- How you avoid overclaiming

Today's lab

Lab goal: write your threat model, limitation, and recommendation

- Restate your project question
- Complete a mini threat model
- Think through evasion and error
- Draft one limitation
- Draft one recommendation

Exit ticket

Before you leave: defend one recommendation

What can I recommend without overclaiming?

- One recommendation
- One evidence point
- One limitation
- One privacy or ethics tradeoff

Source anchors

How this connects to security practice

- NIST describes adversarial machine learning attacks and mitigations
- OWASP names data poisoning as a machine learning security risk
- Risk management means naming limits, not promising perfection

Work time

Use the rest of class to strengthen your final project

- Open the Day 2 lab handout
- Write the limitation before you forget it
- Make the recommendation specific
- Submit by the end of class